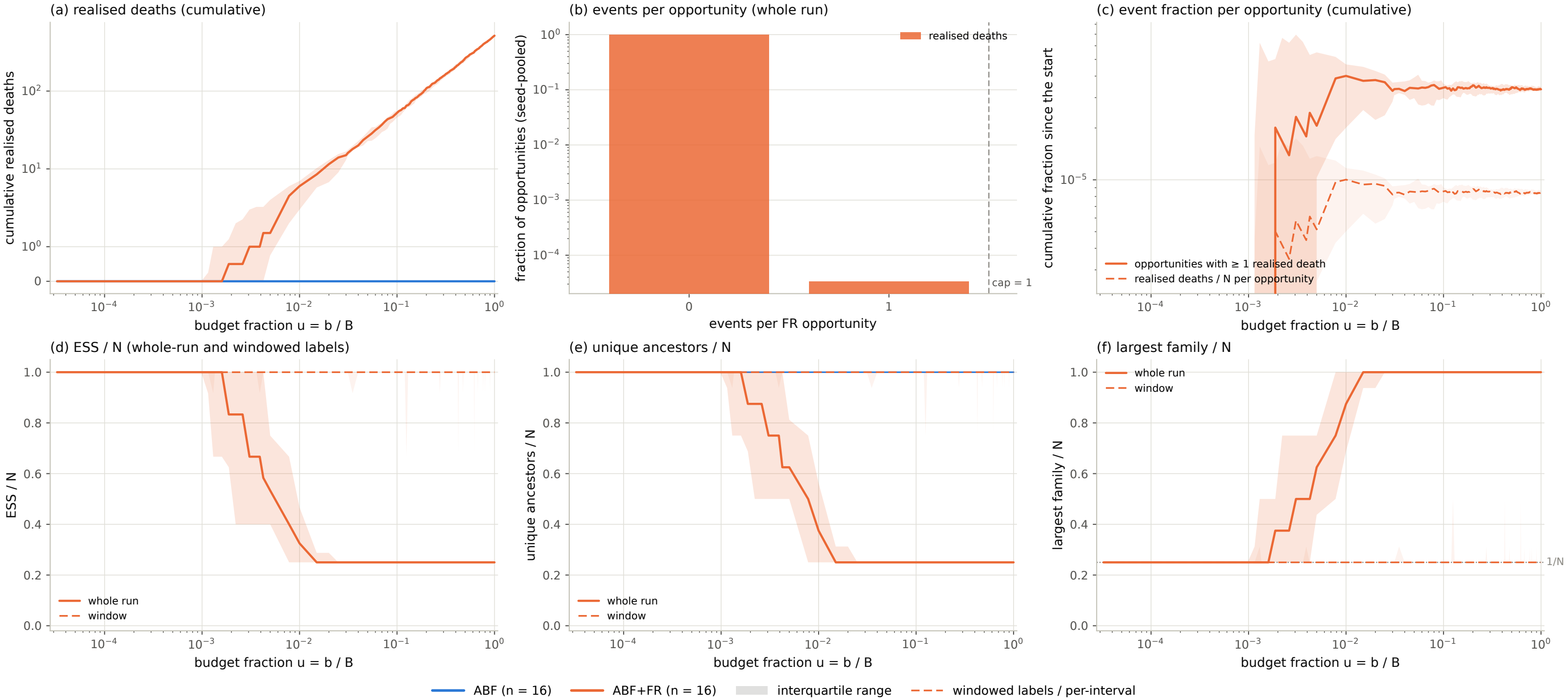


Ethane/LTA 300 K · N = 4 · T_N = 15360 t.u.

n_steps = 76,800,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

FR genealogy and activity vs budget fraction u



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

FR cap = 1 events per opportunity (finite-N cap extension $\max(1, \text{floor}(0.02 N))$): the historical $\text{int}(0.02 N)$ is 0 here); genealogy window 4.0 t.u.; ABF: ESS and family sizes undefined (each walker its own lineage), unique ancestors = N (blue line in e).

Events per opportunity: realised deaths (one death + one copy each); seed-pooled whole-run histogram, log scale.

N < 50: uses the finite-N cap extension (SCIENTIFIC_PLAN section 1).